Electrophysiology-Based Closed Loop Optogenetic Brain Stimulation Devices: Recent Developments and Future Prospects

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(Methodological Review)

***Abstract*—With its potential of single cell specificity, optogenetics has made the investigation into the brain circuits more controllable. Closed loop optogenetic brain stimulation enhances the efficacy of the stimulation by adjusting the stimulation parameters based on direct feedback from the target area of the brain. It combines the principles of genetics, physiology, electrical engineering, optics, signal processing and control theory to create an efficient brain stimulation system. To read the underlying neuronal condition from the electrical activity of neurons, a sensor, sensor interface circuit, and signal conditioning are needed. Also, efficient feature extraction, classification, and control algorithms should be in place to interpret and use the sensed data for closing the feedback loop. Finally, a stimulation circuitry is required to effectively control a light source to deliver light based stimulation according to the feedback signal. Thus, the backbone to a functioning closed loop optogenetic brain stimulation device is a well-built electronic circuitry for sensing and processing of brain signals, running efficient signal processing and control algorithm, and delivering timed light stimulations. This paper presents a review of electronic and software concepts and components used in recent closed-loop optogenetic devices based on neuro-electrophysiological reading and an outlook on the future design possibilities with the aim of providing a compact and easy reference for developing closed loop optogenetic brain stimulation devices.**

***Index Terms*—Closed-loop optogenetics, neurosignal acquisition, neurosignal processing, neurostimulation, phase-locked neurostimulation.**

I. Introduction

OVER the past three decades, brain stimulation techniques have evolved into powerful tools for investigating brain functions and as treatment options for different brain conditions [1] including Parkinson's disease (PD), Alzheimer's disease (AD), multiple sclerosis (MS), epilepsy, and Major Depressive Disorder (MDD. Among all the brain stimulation methodologies, Deep Brain Stimulation (DBS) has gained much popularity. DBS is a highly invasive neurosurgical procedure which works in a similar way as artificial cardiac pacemakers. While a cardiac pacemaker maintains suitable heart rate by electrically stimulating the heart muscles, a DBS device delivers electrical stimulation to the target area of the brain using electrodes [2]. The electrodes are invasively implanted into the brain for this purpose. This electrical stimulation helps control the neural activity in the target brain region.

Along with the developments in DBS techniques, non-invasive and less expensive modes of brain stimulation modalities are also being considered [3]. Transcranial magnetic stimulation (TMS) is one of those stimulation techniques which is non-invasive, where stimulation is achieved using pulsed magnetic field through electromagnetic induction [4]. TMS is an approved treatment option for conditions like MDD and obsessive compulsive disorder [5]. Transcranial electrical stimulation (tES) is yet another non-invasive brain stimulation where electrical current is applied to the scalp for neuronal stimulation. Single-pulse, paired-pulse, and repetitive TMS (rTMS) are the different types of TMS depending on the specifications of the pulsed magnetic field [6]. Transcranial direct current stimulation (tDCS), transcranial alternating current stimulation (tACS), transcranial random noise stimulation (tRNS), and transcranial pulsed current stimulation (tPCS) [7], [8] are the types of tES methodologies under research and clinical trials, as listed in Fig. 1.

With DBS and other non-invasive brain stimulation modalities, the exact neuronal circuits responsible for the therapeutic effects are unknown as it is not easy to control the specificity of stimulation [9]. Accordingly, to elucidate the underlying mechanism behind the occurrence of disease condition and dissecting the role of different neuronal circuitry, researchers have focused on optical brain stimulation called optogenetics [10].

As the term suggests, optogenetics involves the incorporation of concepts of optics and genetics to manipulate well-defined events within specific cells of living tissue. The light insensitive nerve cells are made sensitive to light by genetically transfecting them to express light sensitive protein molecules. Upon light illumination, these light sensitive protein channels open and facilitate the movement of ions across the membrane causing the change of membrane voltage potential. As a result, the neuronal cell undergoes depolarization and thus the activity of the genetically modified neurons can be precisely controlled.

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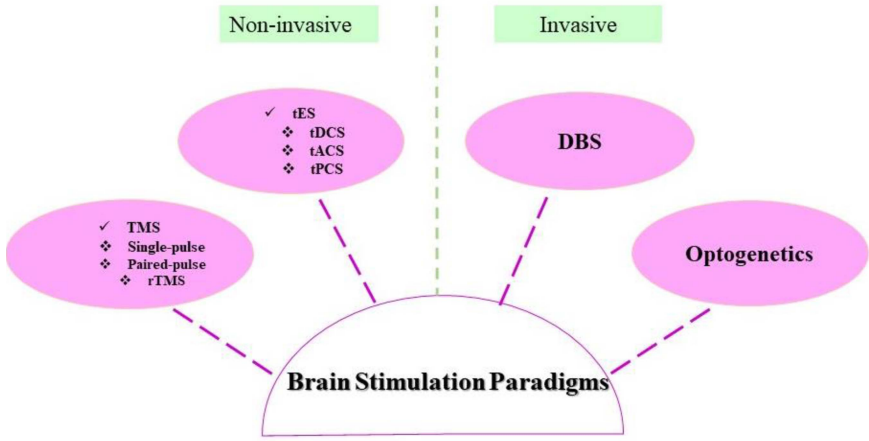


Fig. 1. Types of brain stimulation methodologies.

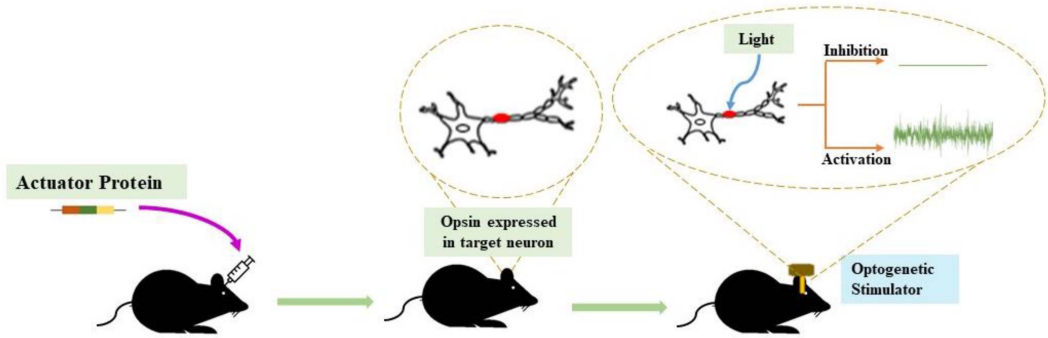


Fig. 2. Processes in optogenetic brain stimulation in transgenic mice – introducing optogenetic actuator protein in the target neuron followed by stimulation using light of specific wavelength to cause activation/inhibition of the target neuron.

Thus, optogenetics offers the ability to control one type of cell, while leaving others unaltered.

The fundamental requirement in optogenetic brain stimulation is the introduction of the optogenetic actuator proteins [11] to the neurons, using genetic engineering techniques. These are the proteins that makes the cell responsive to light by altering the electrical activity of the cell in which they are introduced when exposed to light of a definite wavelength. Opsin is the widely used optogenetic actuator protein in various studies. Opsins are the light sensitive proteins that absorb light at specific wavelengths [12]. Upon exposure to light of particular wavelength and intensity, these channels respond by opening or closing allowing a flux of ions into or out of the cell. Opsins underlie the molecular basis of various light-sensing systems including phototaxis, circadian rhythms, eyesight, and certain types of photosynthesis. Commonly used optogenetic opsins are types I (microbial opsins) and II (animal/vertebrate opsins). Microbial opsins are single component light sensing systems that offer fast and precise manipulation of neural excitability by directly altering the membrane potential of a neuron to bring it closer to, above, or further away from the threshold for generating action potential. Opsin is expressed in the neural population by either virus-based transfection method, or by developing transgenic mice line. The commonly used opsins in optogenetic studies, channelrhodopsin (ChR), and halorhodopsin (HR) are microbial opsins. By expressing ChR, a cation channel activated by light, neurons will be activated by delivering blue light at required specifications. While, HR - an inhibitory opsin, produces inward chloride flux resulting in the inhibition or silencing of neuron on stimulation [13]. Fig. 2 illustrates the process of optogenetic brain stimulation.

With the cell specificity, optogenetic brain stimulation (OBS) can deliver optical stimuli that are more focused than other brain stimulation approaches [9]. This allows manipulation of specific neuronal circuits, and can help in better understanding of how different neuronal circuits get altered under different neurological disease conditions [14]. Apart from its specificity, another advantage of optogenetic brain stimulation is the ability to excite or inhibit neural activity with the co-expression of excitatory and inhibitory opsins. This enables the bi-directional control of neuronal circuits with lights of different wavelengths [15]. The direct inhibition of the neurons with inhibitory opsins have important clinical applications in treating neurological conditions such as seizure, and acute pain [16].

On the device engineering side, the control strategy used in most of the commercially available brain stimulation systems, irrespective of the methodology, is open loop [17]. This is an important issue to be addressed. One of the main characteristics of neurological disorders is that the associated symptoms do not stay constant. The conditions of the patients vary throughout a period. Because of the open loop nature of the control strategy, patients may receive more or less stimulation doses than their

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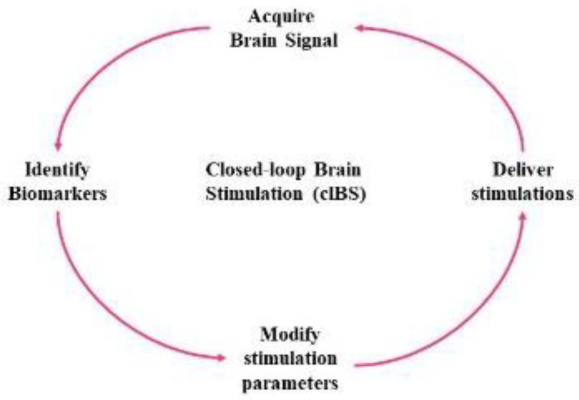


Fig. 3. Steps in Closed-loop Brain stimulation.

symptoms require. Too much stimulation can cause side effects, while too little could mean symptoms are not well managed. In addition, extra stimulation uses up battery life unnecessarily. Moreover, open loop stimulation is not a good choice for diagnostic purposes. A novel approach to brain stimulation to solve these issues is the introduction of closed loop control in the stimulation process [18]–[20], where stimulation parameters can be adjusted based on live neurological conditions, and optimized stimulation is provided. Literature indicates that closed loop brain stimulation (clBS) has offered promising results in pain relief [21], [22], epilepsy seizure reduction [23], [24], and motor response improvement in PD [25], [26] 42, 43. Closed loop brain stimulation paradigms, as given in Fig. 3, use brain signals, called biomarkers, as a feedback signal to modulate the stimulation in real-time. Using extra circuitry, the biomarkers are quantitatively analyzed to give precise stimulation at accurate times.

Accordingly, the closed loop optogenetic stimulation can provide a distinctive method for efficient cell-type targeting and controlled neuromodulation. Currently, closed loop optogenetic stimulation is used to study neural circuits that underlie both healthy and pathological conditions. This information is then used to guide the development of effective diagnostics and therapeutics. And recently, a closed-loop DBS device from Medtronics has been approved by the United States Food and Drug Admiration for treatment of PD symptoms [27].

An overview of a closed loop optogenetic brain stimulation system with brief description of the components is presented in the next section.

II. Closed Loop Optogenetic Brain Stimulation

The essential difference between closed loop and open loop control systems lies in the feedback mechanism. In a closed loop control system, a feedback control is in action to modulate the variables of the system depending on the actual output and the desired output. The feedback signal, thus, aims to bring the output of the system to the desired value. The same is the case with the optogenetic brain stimulation system as well. The functional block diagrams of an open loop and a closed loop optogenetic stimulation system are presented in Fig. 4. In an open loop OBS system, the stimulation parameters (stimulator specifications) are changed manually based on the user feedback. Because of the manual nature of the control and the uncertainty in the neural conditions of the patient, it often takes frequent hospital visits to control the stimulation effectively. In closed loop optogenetic stimulation systems, however, the stimulation parameters are modified automatically with the help of the feedback signal from the controller. The feedback signal is obtained by sensing and analyzing different measurable medical signs, called biomarkers [28]. Electrophysiological and neurochemical signals are some of the measurable biological signals that can be used to get an understanding of the underlying state of the neuronal network [29].

Biomarkers are sensed using electrodes implanted into the patient's brain. A sensor circuit, comprising of amplifiers and filters, conditions the sensed signal for further analysis. The controller extracts the features of the sensed biomarkers, classifies the biomarkers, and generates control signal for closing the feedback loop. A control algorithm identifies deviations, if any, of the actual stimulation output from the required level of stimulation. Accordingly, the algorithm adjusts the stimulation parameters to minimize the difference in actual and desired outputs with the help of the stimulation control circuit. Thus, feedback control helps make the brain stimulation more efficient. Also, as the stimulation parameters are not modified manually, thus frequent manual interrupts would not be required. This helps improve the diagnostic efficiency of the closed loop optogenetic stimulation systems by enabling uninterrupted study with freely behaving subjects. The essential stages in a closed loop optogenetic brain stimulation can be listed as: neural sensing, feature extraction, classification, control, and optical stimulation.

To fully leverage the closed-loop scheme, it is an important consideration for the closed loop optogenetic brain stimulation (clOBS) system to be untethered [30]. Physically tethered technologies such as those using power cables and optical fibers can pose difficulty for freely behaving animals as the tethered systems will affect the free mobility and behavior of the subject under study. Tetherless operation is enabled in optogenetic systems by using small implantable light sources, and through replacing wired power supply with small and light-weight batteries. Along with this, in many of the recent publications on portable clOBS systems, a transceiver is also included which transmits the digitized signals from the sensor to a nearby tethered system with higher computational capabilities and power requirements. In such a system, the optogenetic system receives control signals from the nearby tethered system. Wireless power transmission techniques using inductive or capacitive coupling can also be seen in some of the recent systems [31].

The following sections describe each of the components (biomarkers, biomarker sensing circuitry, stimulation circuitry, and signal processing and control) of untethered closed loop optogenetic brain stimulation devices in detail, and review the recent literature to identify the methods in which these are implemented in the recent works.

III. Biomarkers: Signals and Features

Biomarkers are measurable signs that indicate the physiological as well as pathological conditions of an individual subject. Neurons communicate with each other using electrical action potentials (formed by transfer of charged ions across synapses)

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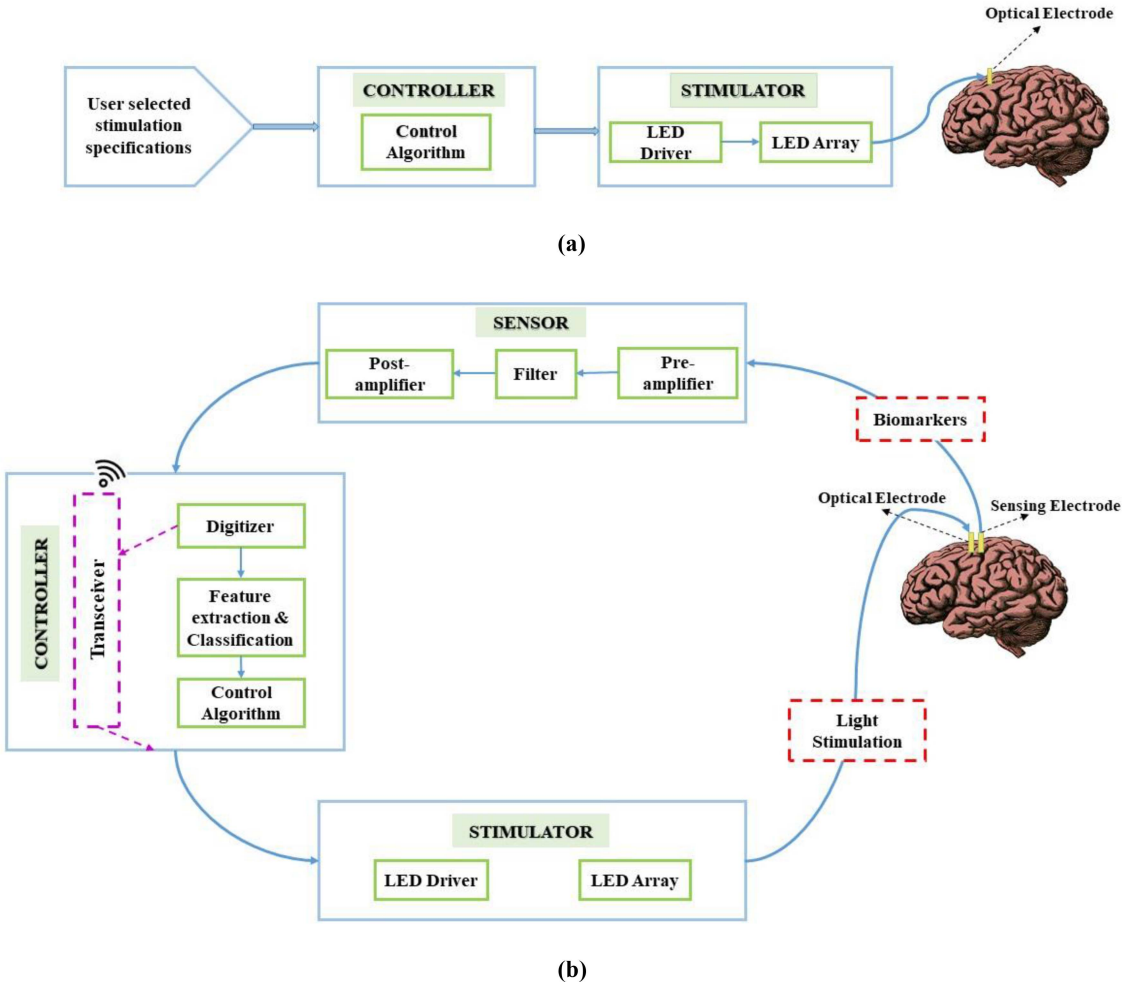


Fig. 4. Functional block diagrams. (a) An open loop optogenetic stimulation system. (b) A closed loop optogenetic stimulation system.

and chemical neurotransmitters. Hence, the essential biomarkers for studying brain and diagnosing and monitoring neurological pathologies can be obtained from the measured bioelectric (electrophysiological) and biochemical signals. A biochemical brain signal refers to the neurotransmitters which are considered as the chemical messengers. Some of the key neurotransmitters are dopamine, glutamate, gamma aminobutyric acid (GABA), glycine, serotonin, epinephrine, endorphins, and acetylcholine, to list a few. Normal brain function requires a precise balance of chemical signaling done by the release of specific neurotransmitters. Important body functions like sleep, memory, metabolism, temperature control, mood, motor control etc. are closely associated with neurotransmitters [32], and hence the level of different neurotransmitters in the measured biochemical signals can function as the biomarkers for many neurological pathologies including PD [33], AD [34], MDD [35] etc. On a similar note, different wave characteristics like amplitude, power, frequency, etc. of the electrophysiological signals can serve as the diagnostic measure of different neurological conditions [36]–[38]. For using these signals for closing the loop in brain stimulation systems, efficient measurement is required.

For monitoring changes in neurotransmitters, different methods have been employed. Fast-Scan Cyclic Voltammetry (FSCV), high speed chronoamperometry, and potentiometry, impedance spectrometry are some of such methods [39]. Among these, FSCV has been a popular method used in some of the recent deep brain stimulation studies. For example in ref. [40], the antidepressant effect of the medial forebrain bundle neural pathway is studied by combining DBS with FSCV to measure dopamine concentration in the ventromedial striatum. In ref. [41], a miniaturized FSCV device with a dimension of 40 mm $\times$ 24 mm is reported which enables portability and use in tether-less closed loop settings.

Electrophysiological signals acquired using electroencephalography (EEG), electromyography (EMG), electrocorticography (ECoG), and invasive microelectrode arrays (Local field potentials (LFPs), Action Potentials (APs)) can be used as a feedback signal in closed-loop brain stimulation systems. Each of these signals has different method of recording, level of invasiveness, information content, resolution, and medical significance [42]. Fig. 5 summarizes the main features of the electrophysiological signals. EMG records myoelectric signals which are formed by the changes in the state of muscle fiber membrane [43]. EEG contains the recording of the electrical activity of cortical neurons over a short period of time using electrodes placed on the scalp [44]. Spatial resolution of EEG

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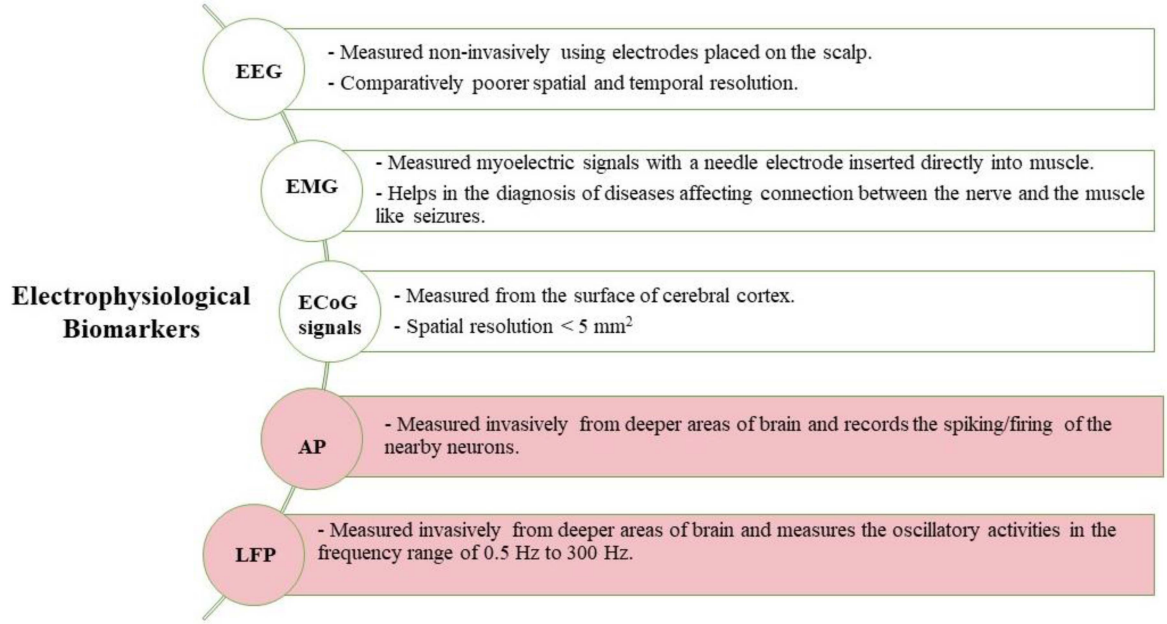


Fig. 5. Main features of electrophysiological biomarkers.

depends on the number of electrodes used, even with more than 200 electrodes, the spatial resolution is still low [45]. In ref. [46], a spatial resolution of 22-37 cm^3 is reported for EEG measured with 19 scalp electrodes, and 6-8 cm^3 with 129 electrodes. ECoG or intracranial EEG (iEEG) is invasive and uses subdural electrodes placed on the surface of cerebral cortex to measure the cortical neuron activity and is less susceptible to artefacts when compared with EEG [47]. ECoG has a comparatively higher spatial resolution of $<$ 1 cm and temporal resolution of $<$ 1ms [48]. In contrast to EEG and ECoG, LFP and AP are the electrophysiological signals that captures the neural activity from the deeper locations of brain. These are the extracellular signals acquired using metal or glass microelectrodes or microelectrode arrays. LFP and AP signals offer better spatial resolution, less distortion and attenuation from tissues owing to its small contact area [42]. Much progress has been attained in acquiring these LFP and AP signals with advancements in microelectrodes design.

IV. Action Potential and Local Field Potential

One of the major advantages of using microelectrodes and microelectrode arrays, with tip dimension of the order of micrometers, is the opportunity for size reduction and realization of untethered brain stimulation devices. As mentioned earlier, the recorded electrophysiological signal picks up two components – AP and LFP. The AP, which are extensively studied encodes the spiking or firing of the nearby neurons. Whereas, LFP oscillations are the summation of variations in synaptic potential across thousands of neurons near the electrode from a volume of up to several hundred cubic micrometers [49]. The acquired signals are low pass filtered at around 300 Hz cut-off frequency to obtain the LFP frequencies. The oscillatory activities of different frequency bands in the LFP signal (delta band – 0.5 to 4 Hz, theta band – 4 to 8 Hz, alpha band – 8 to 15 Hz, beta band – 15 to 30 Hz, gamma – 30 to 300 Hz) are the sources of information about the underlying brain pathologies, and studies indicate that multiple brain oscillations could be altered in different pathological conditions of brain. Alterations in oscillation amplitude, phase, and coherence, which are studied by spectral analysis of the acquired signal, are some of the indications of underlying neurological pathologies [50]. For example, excessive activity in beta band is linked to movement disorders in PD [51]. In a similar study, Swann *et al.* [52] proposes hyperkinetic disorders in PD is connected to gamma band activities in neural oscillations. Neumann *et al.* [53] proposes alpha band activity as a biomarker for MDD symptom severity following the studies showing elevation in alpha power in the human limbic system in patients with MDD. The role of slow gamma oscillations (30-60 Hz) in memory in AD is established in [54]. References [55], [56] reported aberrant gamma band oscillations in Schizophrenia. Recent closed-loop optogenetic brain stimulation experiments and systems involving electrophysiological biomarker sensing is reported in Table I.

In this work, we focus on reviewing electrophysiological signal based closed loop optogenetic brain stimulation systems with an emphasis on the requirements of untethered operation. The remaining sections of the paper focuses on concepts and components for the realization of such systems. The following section discusses the circuits for electrophysiological biomarkers sensing circuits in recent brain stimulation systems.

V. Biomarker Acquisition and Sensing

Untethered closed loop optogenetic stimulation demands an implantable neural probe for combined light stimulation and electrophysiological signal measurement. Such neural probes are referred to as optrodes [64], [65]. The rapid developments

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TABLE I
RECENT CLOSED-LOOP OPTOGENETIC BRAIN STIMULATION SYSTEMS WITH ELECTROPHYSIOLOGICAL BIOMARKER SENSING

Reference	Signal/ Technology	Signal Specifications	Purpose
Siegle et al. [57] (2014)	LFP	Theta oscillations (4-12 Hz)	Hippocampus dependent memory
Magloire et al. [58] (2019)	iEEG/ ECoG	Spike Amplitude	Epileptic seizure
Kanta et al. [59] (2019)	LFP	Gamma Oscillations (30 – 100 Hz)	Spatial memory consolidation
Sorokin et al. [60] (2017)	iEEG/ ECoG	Line length threshold	Absence epilepsy
Streng et al. [61] (2020)	LFP	Spike amplitude, width, frequency, deflection	Temporal lobe epilepsy
Paz et al. [62] (2013)	EEG	Line length threshold	Epileptic seizure
Latchoumane et al. [63] (2017)	LFP and EEG	Sleep spindles (12 -14 Hz)	Hippocampus dependent memory

in the microelectronics industry supported the development of small implantable optrodes with multiple stimulation and recording sites on board. However, the stimulating light falling on the metal electrodes can generate electrical artefacts in the recorded signal. The artefacts are not easy to distinguish from the neural signal, especially the low frequency LFP signals. And to avoid this, recently transparent graphene electrodes are used in some of the closed loop optogenetic systems as careful design of the important fabrication process steps for transparent graphene electrodes can lessen the light-induced artefact problems [66]–[69].

Once the bio-signals are acquired using microelectrodes, the next step is to process them to extract information of the underlying brain condition of the subject. Often the acquired signals will be very feeble (with amplitudes in μV and a few mV range), with varying frequency, and highly corrupted by external and device noise. Moreover, in the case of closed loop brain stimulation, due to the closeness of stimulation and recording electrodes in the neural probes, artefacts can distort the signals. Thus, in order to perform efficient digital processing of the signals, it is essential to condition the received analog signals prior to digitization. Analog signal processing of the acquired signal is thus done using the analog front end (AFE) circuit or a sensor circuit which consists of filters and amplifiers to amplify the low-level signals and filter out the unwanted frequency components including the noise. Accordingly, the favorable features of a signal conditioning circuitry are: firstly, high gain, high input impedance amplifier with suitable bandwidth (BW), high signal-to-noise ratio (SNR), common-mode rejection ratio (CMRR) and power supply rejection ratio (PSRR) to boost the weak analog bio-signals, and secondly, an effective filtering to remove any unwanted frequency components in the signals corresponding to the artefacts and noise, like the large DC offset causing from the electrode tissue interface [70]. Finally, the conditioned signals are then used to drive the analog to digital convertor. As expected from any stand-alone, wireless implantable system, low-power performance is anticipated from the circuit.

Over the years, several AFE circuits for recording neural signal recording interfaces have been developed for various untethered closed-loop brain stimulation systems. Both application specific integrated circuits (ASIC), and commercial off-the-shelf (COTS) components based on microcontroller units (MCU) and field programmable gate arrays (FPGAs) are adopted in designing AFE in the closed loop systems. Fig. 6 shows the block diagrams of the neural sensors in recent clBS systems discussed in this section and Table II compares the neural sensor specifications in those systems.

In some of the early untethered closed loop brain stimulation systems commercially available digital electrophysiology interface chips are used. Liu et.al. [71] presented a closed loop optogenetic brain stimulation system for behaving animals with real time electrophysiological recording. The electrodes are directly connected to a digital electrophysiology interface chip (Intan RHD2216) using a ZIF connector. The chip contains an array of low noise amplifiers with programmable bandwidth (250 Hz - 7.5 kHz), analog and digital filters, and 16-bit ADC. The interface chip is controlled by a low power microcontroller (MSP430). The microcontroller drives the interface chip to digitize the analog signal (sampling rate of 30 kHz/channel). Following this, the digitized signal is directed to the MSP430 through the SPI interface to realize feature extraction, and closed loop control operation. The electrophysiology system has a dimension of 31.8 mm × 21.8 mm. A similar approach is used in the LFP sensing circuit for the wireless head stage in ref. [72] using the Intan RHD2132 neural recorder.

Further reduction in form factor for better untethered operation was realized by using commercially available instrumentation amplifier and operational amplifier ICs. Jia *et al.* [73] reported a 2-channel neural interface for recording ECoG signals for wireless optogenetic brain stimulation. Here the stimulation parameters are controlled through a graphical user interface (GUI) after analysing the ECoG signals. The circuit has active high pass filter made of the commercially available low power, high precision instrumentation amplifier IC, INA118 from Texas Instruments. The output from INA118 is used to drive an active low pass filter stage realized using low-noise, AD8603 from Analog Devices, and other passive components. The low pass filter output is then subtracted from the input to provide a high pass filtered output. Finally, an 11-bit built in ADC in CC2441 microcontroller is used to digitize the signal. The input referred noise of the channels are 8.47 μV_{rms} and 8.64 μV_{rms} and has a total

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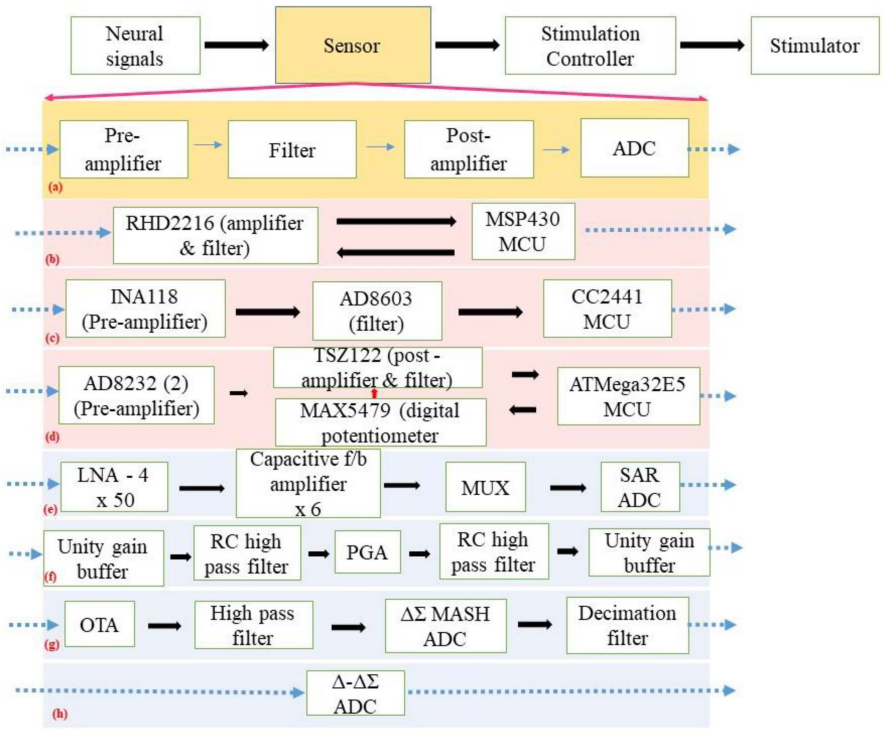


Fig. 6. Block diagrams of neural sensors in recent clBS systems. (a) Hardware components in a neural sensor. (b) Block diagram of the neural sensor in [71]. (c) Block diagram of the neural sensor in [73]. (d) Block diagram of the neural sensor in [74.] (e) Block diagram of the neural sensor in [75]. (f) Block diagram of the neural sensor in [77], [78]. (g) Block diagram of the neural sensor in [79].

power consumption of 1.5 μW per channel. The processed and digitized data is sent to a user interface via a Bluetooth link for monitoring the state of the brain and produce the control signal.

Parastarfeizabadi *et al.* [74] presented a 2 channel COTS-components based neural interface design for wireless closed loop DBS. The device has the capability of recording both AP and LFP signals. The device consisits of four parts – a pre-amplifier with a gain of 100 V/V, which is realised using a low-noise, low-offset, high common-mode rejection ratio precision instrumentation amplifier IC AD8232. Filtering and post-amplification is realised using TSZ124, a high accuracy quad operational amplifer IC, and other passive components. The post amplification is made programmable (1-1000 V/V) by including a digital potentiometer chip, MAX5479. The digital potentimeter is programmed using ATMega32E5 MCU. The amplified neural signals are directed to the MCU for digitization using its inbuild ADC.

While COTS offers low cost sensor design, important device parameters like number of input channels, power consumption, and noise rejection factor are not on-par with the ASIC based sensors. Chen *et al.* [75] presented a monolithic IC integrating a neural amplifier for combined electrophysiological recording and laser/LED driver for optogenetic stimulation using 0.18 μm CMOS processing technology. The first amplifying stage is a unity gain buffer to form a high input impedence stage to improve the SNR. A high pass filter is formed using capacitors and transistors to eliminate the DC offset. A low-noise, low-power operational transconductance programmable gain amplifier (PGA) forms the following stage. The gain is adjustable by modifying the ratio of the external resistances. A second stage RC high pass filter formed by a capacitor and two metal–oxide–semiconductor field-effect (MOSFET) transistors working as variable resistors follow. The subsequent amplifier has an overall gain of 51.5 dB and amplifies the neural signal to a voltage of 200 mV peak-peak. The final stage is a unity gain buffer which will futher feed an ADC for signal digitization. The IC, however, lacks an inbuilt ADC and uses a data acquisition card for digitization. In- vivo and simulation based experimental evaluation were undertaken to characterise and confirm the electrical properties of the neural amplifier and is documented to have a good SNR of at least 6.6, adequate for extracellular in-vivo neural recording.

Ramezani *et al.* [76] presented an ASIC implementation (0.35 μm technology) of an opto-electro neural interface. This work realized a system with 4 LFP recording channels and two stage amplification for each channel. The low noise amplifier (LNA) has a capacitively coupled input with capacitive feedback. This is done to fulfil the requirement of high input impedance (of the order of 10 -100 kΩ). A capacitance of approximately 3.4 pF is used for this purpose. The amplifier uses a differential folded cascode operational transconductance topology with resistive source degeneration for noise suppression and provides a gain of 50. The low pass cut-off frequency is 7 kHz. There is a second stage amplifier to provide enough signal amplification for the analog-to-digital convertor (ADC) requirement. A single ended amplifier with capacitive feedback is used to provide a gain of 6. Further, the outputs forming all the four different channels are multiplexed (controlled by the signal generated by the recording control block) and digitized using a custom-made ADC based on 9:1 fully differential successive approximation register (SAR) with 10-bit resolution.

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TABLE II
ANALOG FRONT END CIRCUITS IN EXISTING CLOSED LOOP BRAIN STIMULATION SYSTEMS

Reference, year	Design type	Method of stimulation	Targeted signals	No: of channels	Gain (dB)	Input referred noise (μV_{rms})	Bandwidth	ADC	Power/Channel
Chen et al. [75] 2017	ASIC	Optical	AP	1	51.5	4.57	5 kHz	NG	NG
Gagnon et al. [77, 78] 2019	ASIC	Optical	AP/LFP	10	-	3.2	0.1 – 7kHz	Delta-sigma modulator + decimation filter	11.2 µW
Luo et al., Ramezani et al. [76, 80] 2020, 2018	ASIC	Optical	LFP	4	49.54	2.08	0.1 – 300 Hz	10 bit SAR	NG
Kassiri et al. [79] 2017	ASIC	Electrical	LFP	64	-	1.13	0.01 – 500 Hz	Delta-sigma modulator + decimation filter	0.63 µW
Zhou et al. [81] [82] 2019	ASIC	Electrical	LFP	128	-	0.071	Up to 500 Hz	15 bit SAR	8 µW
Jia et al.[73] 2018	COTS	Optical	iEEG (ECoG signals)	2	49	8.47 channel1 8.64 channel2	1 – 200 Hz	12 bit SAR inbuilt MCU	1.5 mW
Mendrela et al. [72] 2018	COTS	Optical	AP	32	-	-	-	-	-
Liu et al. [71] 2018	COTS	Optical	LFP	16	-	-	-	-	-
Parastarfeizabadi [74, 83] 2018, 2020	COTS	Electrical	AP/LFP	2	100	-	2 – 40 Hz, 300Hz – 6 kHz	-	3 mW

Gagnon *et al.* [77], [78] reported another ASIC implemetation of an opto-electro neural interface using 0.13 µm CMOS processing technology. The single IC design has 10 recording channels, a 3rd order discrete time sigma delta ADC with on chip decimation filter. A differential folded cascade operational trans-conductance amplifier (OTA) forms the first stage of the AFE. The input referred noise of the component is limited to 3.2 $\mu V_{\rm rms}$ using source degeneration resistors. The sensor has the capacility to sense both AP and LFP signals. These multiple modalities are sensed using a tunable high pass cut-off frequency realised by means of the feeback element formed by two capacitors and two pseudoresistor banks to the OTA. Another feature is $\Delta\Sigma$ MASH ADC – 3rd order $\Delta\Sigma$ modulator is followed by a 4th order cascaded integrator-comb decimation filter on the same chip. An improved quantization error and reduced power consumption is reported using $\Delta\Sigma$ MASH ADC instead of the SAR ADC. On a similar note, Kassiri *et al.* [79] reported a 64-channel ASIC design (0.13 µm CMOS processing technology) neural sensor for LFP signals for electrical DBS. The amplification and digitization of the input neural signal is done using a $\Delta\Sigma$ based ADC. The design is $\Delta^2\Sigma$ ADC, with an additional Δ stage for removing input DC offset by implementing a feedback loop to find the temporal difference between successive signals.

As compared in Table II, the custom ASIC design provides better noise rejection factor in terms of noise rms value. The ASIC based design also ensures accommodating more input channels in the neural sensor. Also, the custom made neural sensor design ensures less power consumption than that designed using COTS components, which facilitates more efficient untethered operation. However, as explained earlier, apart from being less expensive, COTS based neural sensor design also has the advantage of easy re-design option.

Once the neural signal is digitized, the subsequent step is to analyze the signal to gain information about the neural state of the subject and generate control signals for modifying the

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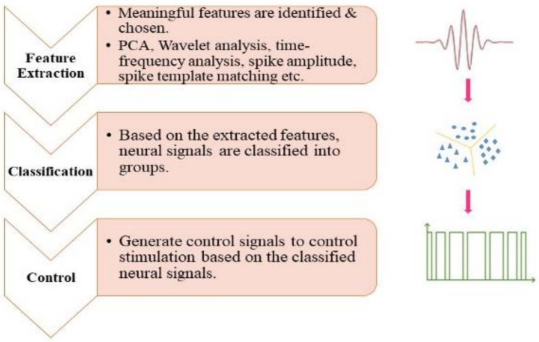


Fig. 7. Algorithm flow in a CIBS system.

stimulation parameters. This is done using digital signal processing methods, and implementing feature extraction, signal classification, and control algorithms. The next section discusses the software components and their deployment on hardware platforms in closed loop brain stimulation systems.

VI. Digital Signal Processing and Analysis: Feature Extraction, Classification, and Control Signal Generation

The purpose of feature extraction is to reduce the number of variables involved in describing the data in order to improve the computational efficiency, and increase processing speed. Features that are meaningful in the classification stage are identified and chosen. This process excludes the outliers and noise from the incoming data. It is followed by classification of the signals based on the extracted features. Finally, based on the classification step, control signals are generated to close the feedback loop of the stimulation process. This algorithm flow in a clBS is illustrated in Fig. 7.

The digitized signal can be either AP (neural spike data) or LFP (neural oscillations). Neural spike data analysis involves spike sorting. Spike sorting consists of the following steps: spike detection of the filtered signal, feature extraction, and clustering. Spike detection is usually done by employing amplitude thresholding on the filtered data. From the detected spikes, critical features are extracted in the feature extraction step. Cutting edge feature extraction approaches like principal component analysis (PCA), wavelet analysis, singular value decomposition (SVD) etc. are also used apart from simple approaches like classification based on spike amplitude and spike template matching [84]–[88]. PCA is a very popular unsupervised learning algorithm, and it takes the input and reduces its dimensionality by finding out the combination of the input features which best summarizes the original dimensions [89].

Discussing about the feature extraction and classification of LFP signals, classically, LFP data analysis involves frequency domain analysis such as power spectral density (PSD) analysis to determine the energy related to the frequency bands of LFP signals [90]. However, owing to the non-stationary nature of LFP signals, time-frequency analysis techniques like short-time Fourier transform, wavelet transform, Hilbert transform etc. best suit the signal for finding information on phase, frequency, and temporal features.

After feature extraction, the signals are classified into groups. Binary classification is a commonly used method in portable clBS, where the neural signals are classified into either normal or abnormal signals. More sophisticated machine learning (ML) based classification algorithms [91] are computationally studied in classification of tremor, and they could improve the efficacy of the classification in clBS systems. Some of the studies reported the use of artificial neural networks (ANNs) in classification of tremor in PD from wearable sensor signals [92] and EMG signals [93]. In ref. [94], ε-recurrence networks are adopted to classify resting tremors in PD. In [95], a computational study of a two-stage neural network is done for classification of sleep stages from EEG and EMG data with a reported accuracy of 80.6%.

The most commonly used control strategy in portable closed loop optogenetic brain stimulation devices, or generally in closed-loop neuroscience, is the ON/OFF control, where the stimulation is turned on and off depending on the events extracted from the processed bio-signal. The strategy commonly employed is to compare the detected activity against a preset threshold value/template and turn the stimulation ON/OFF accordingly. Apart from that dual threshold is adopted in ref. [96], where the electrical stimulation amplitude can be increased or decreased to keep the sensed biomarker within a specific range. Another control strategy is proportional approach, where the stimulation amplitude is proportional to the measured biomarker. A fuzzy logic based classifier and controller is implemented in the multi disease deep brain stimulation device in ref. [83].

While there are several different approaches used for feature extraction and classification of recorded neural signals, applying many of these algorithms to untethered implantable devices is a challenging task. The computational limitation imposed by wireless battery powered platforms like MCU, FPGAs, and DSP is the bottleneck [97]. Hence, some sophisticated methods of neural signal analysis for feature extraction and classification are often implemented offline or by sending the data to an external computer. For example, while realizing a closed loop optogenetic system for studying temporal lobe epilepsy, Krook *et al.* [98] and Armstrong *et al.* [99] extracted information on the occurrence of seizure in epileptic mice subject by analyzing the EEG using spike sorting algorithm running on a computer. The amplified and digitized hippocampal EEG signal is relayed to a computer, where a custom-designed software (created in MATLAB) was used to detect seizures real time. The algorithm uses signal power properties, spike features, and frequency properties. Triggering is initiated in the following conditions: (i) Amplitude correlation trigger: Amplitude correlation of the signal of the current frame (0.12s) with the previous frame is done, and triggering is performed if the correlation is greater than a set threshold value; (ii) Narrow spike trigger: Triggering is enabled when the number of narrow spikes exceeds a set threshold value during the time frame; (iii) Spike ratio trigger: Here if the distance of spike train exceeds the minimum number specified by the user, triggering is enabled; and (iv) Frequency band ratio trigger: Triggering is enabled when the relative contribution of the power of specific bands of frequencies to the frequency band of the signal exceeds or drops below a threshold

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value. A simple ON/OFF controller is employed for controlling the stimulation. Following the trigger, light stimulation is made ON in 50% of the cases. In the absence of the trigger, light stimulation is made OFF.

On a similar line, Pashaie *et al.* [100] while realizing a tethered closed-loop rain interface protocol using optogenetics, implemented a proportional control algorithm by real-time processing of the data on a computer. Integral of the incoming ECoG signal is calculated over a 100 ms period. Finally, an error is calculated by taking the difference of the measured voltage signal and the corresponding pre-set desired voltage level for each electrode. The final step involves adjusting the duration of the light stimulation to minimize the error signal. The error in the form of voltage is translated to pulse duration (adjusting the pulse widths) by means of a proportional constant, K_p. The optimal value of the proportional constant is found by using a calibration process in the computer to be used for the realization of the control algorithm.

For untethered systems, hardware platforms such as custom ASICs, FPGAs, and MCUs are being used. FPGAs and MCUs offer portability and configurability. However, due to memory constraints, algorithms deployed on MCUs are rather simple ones. In [71], the sensed and digitized recording is processed in a simple threshold detection algorithm to detect any abnormality in the sensed neural signal. Here, the threshold algorithm runs on-board a standard microcontroller (MSP430 FR59891) in the device. A simple ON/OFF control algorithm is used. If any of the signals sensed from the recording sites exceeds a certain threshold amplitude, which was set as 200 μV, the light stimulation is turned ON, otherwise LED remains in OFF state. More complex feature extraction, disease classification, and control algorithms in portable systems are deployed in FPGAs with more memory resources.

Xia *et al.* [101], [102] used a Flash-FPGA based implementation of seizure detection and untethered closed loop optogenetic stimulation control in rodent. Feature extraction is performed using wavelet filter followed by threshold detection. A proportional-integral-derivative (PID) controller is employed for modulating the pulse width of the stimulating pulses. The proportional, integral, and differential kernels are stored as look up-tables in the flash FPGA hardware. The incoming digitized signal is convolved with these kernels. This is followed by converting the convolution outputs to optogenetic pulse width modulation (PWM) stimulus.

Gagnon *et al.* [103] utilized a low power FPGA for implementing a digital spike detector and a wavelet data compression module in their untethered optogenetic stimulation device for combined optogenetics and multichannel electrophysiology recording. Threshold detection and wavelet data compression is done using an embedded digital signal processing core and a Microblaze soft processor core implemented inside a low-power LX16 Spartan-6 FPGA from Xilinx. The detected and compressed data is transmitted to a computer using a radio transceiver and the stimulation pattern is also received from the computer using the transceiver. The same group developed an ASIC based approach for controlling the stimulation based on AP threshold detection, classification, and triggering based on a sliding window [77]. The FPGA is replaced by an ASIC named as 'neural decoder' core. The ASIC houses a detection core for detecting the spikes based on amplitude thresholding, a compression core for compressing the data using discrete wavelet transform for low-power transmission to a nearby computer, and a classification core for classifying the AP based on wave shape. A closed loop algorithm based on the classified AP is implemented on a low power FPGA (Igloo AGLN250 FPGA). The closed loop algorithm works based on a sliding window, which counts the number of spikes crossing the threshold in a specific duration. The closed loop is implemented by producing a stimulation pattern after a predefined number of threshold crossing occurs within the sliding window.

In a recent work, Luo *et al.* [80] deployed a closed loop algorithm on an embedded processor which is an ARM Cortex-M4 microcontroller with DSP function block. The algorithm is realized in three blocks: a detector module, a suppressor module, and an optical convertor module. The detection module employs three criteria for detecting abnormalities, if any, in the incoming signal. The criteria include measuring amplitude thresholding, changing rate of amplitude over a period of time, and power spectral density. The suppressor module contains a phase shift algorithm and a proportional–integral–derivative (PID) controller. The phase shift algorithm determines the intensity of the signal at a given frequency and then returns a corresponding signal with a phase shift. The optical convertor translates the control signal into the corresponding optical output. There are four modes: (i) threshold maximum – when the incoming signals are above the set threshold, set the output stimulation to be maximum, (ii) pulse amplitude modulation – converts algorithm output into LED digital-to-analog convertor (DAC) values using pulse amplitude modulation, and (iii) pulse width modulation – convert algorithm outputs into LED DAC values using pulse width modulation.

While all of the above mentioned algorithms work based on controlling the stimulation by providing stimulation pulses on the detection of any abnormal signal, there is a demand for closed-loop algorithms to implement another control scheme for suppressing or amplifying the neural oscillations in only the frequency bands of LFP signals. This control approach is called phase-locked stimulation [104], [105]. The rationale behind this control approach is that delivering stimulation pulses at peaks of an oscillation results in constructive interference and resultant enhancement of the ongoing oscillation, whereas delivering stimulation pulses at troughs of an oscillation produces destructive interference and resultant diminishing of the ongoing oscillation. Hence it can be deducted that for enhancing the power of a particular frequency band of neural oscillation, stimulation should be applied in-phase with that oscillation. This theory is supported by some recent research in this field [106], [107], where the power of only a specific frequency band was enhanced by in-phase stimulation. For realizing this control strategy, the algorithm in place should be able to perform accurate and fast estimations of the phase of the LFP data in real-time. The literature on untethered closed-loop phase-locked optogenetic stimulation is very limited. Reviewing the related research in phase estimation algorithm for in-phase neuro-stimulation strategy, in

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an early study, Chen *et al.* [108] developed an algorithm to assess the phase of LFP samples (4-9 Hz) using passband optimization, zero-phase filtering, autoregressive model-based prediction, and Hilbert transform. However, the algorithm produces a delay of around 100 ms between the identified phase and target phase. A continuous phase estimation using autoregressive model for signal prediction with reduced stimulation delay is discussed in [109].

Once feature extraction, classification, and control algorithms are in place, the next step in a closed loop optogenetic brain stimulation system involves the implementation of the stimulation circuitry which gives timely and efficient stimulation pulses based on the control signals.

VII. Stimulation

For a closed loop operation, the stimulation circuitry controls the stimulation light parameters based on the received control signals from the signal processing and control signal generation stage. Frequency, pulse width, and intensity of the stimulating light are important stimulation parameters. Parameters such as the frequency and pulse width can be controlled to give the right stimulation necessary to produce neural responses. The optimal value of stimulation parameters for effective stimulation is still an open question. In ref. [110], the effective parameters of light stimulation are estimated for ChR2 stimulation conditions in experiments with primary hippocampal cultures and the optimal parameters are presented as optimal frequency range of 11-5 Hz, pulse duration of 1-5 ms, and an interval of 10-30 ms between stimulations. Various studies suggest that a light density of 1-10 mW/mm^2 is needed to evoke action potential in the transfected neurons, depending on the type of opsin used [100].

For untethered devices with an integrated head stage, light from μLED sources placed in vivo are mostly found in recent works. To drive the μLEDs, a suitable driver circuit is required to supply constant current to the LEDs. The input to the LED driver circuit is based on the commands from the control algorithm. The driver subsequently produces an output and makes it available to the LED circuitry, to change the intensity or duration of the stimulation accordingly.

Reviewing different LED driving circuitry in optogenetic brain stimulation experiments using μLEDs as the light source, some researchers have used application specific custom-made ICs, while some others have used commercial off-the-self component to build the circuit. In ref. [72], a custom-made IC based on 0.18 μm complementary metal-oxide-semiconductor (CMOS) technology is used to drive the LEDs. A digital-to-analog convertor (DAC) is available, controlled by the SPI based serial input, to drive the μLEDs with 10-bit resolution and a maximum current of 1.023 mA. A level shifting current mirror is made by cascoded positive-channel metal-oxide-semiconductor (PMOS) transistors is used to drive μLEDs up to 5V.

Chen *et al.* [75] presented a custom-made LED/laser driver IC based on 0.18 μm CMOS technology. It has a two-state current control design using programmable PMOS arrays. The first array gives a maximum current of 300 mA and the second gives a maximum of 30 mA. Thus, the circuit delivers a maximum current of 330 mA to drive the light source. Each array has a 4-bit digital input, to which the PWM signal from the previous stage can be provided to adjust the current outputs based on the neuronal condition.

Ramezani *et al.* [76] presented an ASIC design (0.35 μm CMOS technology) for driving CREE DA2432 μLEDs. The design has a DAC, transconductance amplifier (TCA), H-bridge circuit for bi-directional biasing of the LEDs, and the related control logic. The H-bridge configuration biases the LED in forward mode for stimulation in the first stage, and then biases the LED in reverse mode and inverts the generated charge over the tissue, thus increasing the life of the electrodes. PWM information from the previous stage is used in the circuit by writing into memory cells and reading accordingly using finite state machine (FSM) control logic. DAC with 8-bit resolution, current range of 0 - 1.1 mA, and voltage range of 0-3.3 V is followed by a transconductance amplifier to deliver the current pulses to the μLEDs.

Another ASIC based design (0.13 μm CMOS technology) is proposed in ref. [78], [111] for a 4 channel LED driver. Current flowing to the LEDs is controlled by a regulated cascode current source with feedback and ensures that the LED delivers adequate optical power. An external PWM signal controls the driver current. A current reference source generates the reference current depending on the PWM signal. Wide swing current mirrors are present in the design which helps in reducing the voltage drop inside the chip, thus enabling the use of small battery voltages. The system can provide an LED current of up to 20 mA, and an output optical power of 40 mW/mm^2.

Another approach seen in the literature to regulate LED brightness uses PWM signals from microcontrollers. PWM signal based on the feedback control algorithm is used to turn the LED ON and OFF. And the brightness of the LED can be changed accordingly. As many commercially available microcontrollers already have PWM outputs integrated into their architecture, they can easily derive the PWM signal to switch the stimulation ON and OFF and adjust the duty cycle via the program running on the microcontroller. In ref. [71], a μLED chip is soldered on a stretchy cable and driven via the circuit which is controlled by a microcontroller through pulse width modulated signals. The brightness of the μLED changes with the duty cycle of the pulses. In ref. [73], μLEDs are controlled using PWM signals from the microcontroller CC2541 with the driving adjustable current between 5 to 30 mA. The pulse characteristics are: stimulation period of 100 - 900 ms with 100 ms steps, or 1-10 s with 1 s steps and a duty cycle varying between 0.1 to 0.9%.

There are some systems which use commercially available laser diode (LD) chips instead of μLED, which are driven using commercially available LD drivers. For example, in ref. [112], LT1932 LD driver by Analog Devices is used to drive an LD. It operates as a constant current source between 5 and 40 mA based on the PWM signal coming from the feature extraction section.

Power source to the system is another important aspect of the tether-less optogenetic devices. Different power supply options are discussed in the following section.

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VIII. Power Source

Unrestricted power without making the device bulky increases the efficiency of the portable closed loop optogenetic system. Identification of hardware and software systems and practices to reduce the power consumption overhead should be an important design consideration for realization of portable closed loop optogenetic brain stimulation systems. Gagnon *et al.* [78], Jia *et al.* [73], and Liu *et al.* [71] in their fully wireless closed loop optogenetic stimulation devices, used light weight rechargeable Lithium Polymer (LiPo) batteries. A 40 mAh Lithium polymer battery is used to power the ASIC based system in ref. [78] with a power consumption of 11.2 μW/channel. A bulkier 3.7 V 150 mAh battery is used in ref. [71]. In ref. [73], a 3.75 V, 45 mAh battery is used. Commercially available low drop out regulators are used in these designs along with the battery to provide steady supply of power to different components in the head stage.

With optogenetic stimulation experiments on freely moving animals lasting for weeks being held these days [113], reliable and seamless power source without the hustle of replacing the batteries has become a requirement. Ample developments in the field of wireless power transfer (WPT) technology has enabled many of the recent optogenetic devices to replace batteries with efficient energy harvesting modalities to provide stable power to the system. The WPT technology involves a transmitter device which generates a time-varying electromagnetic field which transmits power across space to a receiver device which extract power from the field and supplies it to the load. Near field power transfer, which uses non-radiative waves like radio frequency (RF) waves, covers shorter distance, and uses magnetic field and electric field to transfer the power through inductive or capacitive coupling respectively. Efficient design of the receiver antenna module with minimal power leakage and small form factor is key to developing a fully closed loop device which generates a time-varying electromagnetic field which transmits power across space to a receiver device which extract power from the field and supplies it to the load Near field power transfer, which uses non-radiative waves like radio frequency (RF) waves, covers shorter distance, and uses magnetic field and electric field to transfer the power through inductive or capacitive coupling respectively.

Efficient design of the receiver antenna module with minimal power leakage and small form factor is key to developing a fully closed loop optogenetic system with wireless power transform. Recently, much research is being done in this area [31], [114]–[117]. In ref. [31], Shin *et al.* presents a flexible, miniaturized, and battery-free optogenetic implant with wireless power transfer and communication achieved by near-field inductive coupling (13.56 MHz). The device is placed in subdermal of the subject. The electronic components of the device include the stimulation light source, μLED (465 nm × 220 μm × 50 μm), an indicating LED (1.6 mm × 0.8 mm × 0.75 mm), 25 pF capacitor (0.4 mm × 0.2 mm × 0.22 mm), and a Schotkky diode for smoothening the transferred power. The device stimulation light source is powered and controlled with the help of a loop antenna interface to an RF generator with 12 W output power. Reliable power transfer is assured within a range of 30 cm.

IX. Discussion and Future Prospects

To fully leverage closed loop optogenetic brain stimulation, it is desirable to have compact, fully implantable, and untethered devices for use with freely behaving subjects. From an engineering point of view, recent research on optogenetic brain stimulation focusses on developing miniaturized untethered devices. It mainly focuses on using commercially available components as well as custom made ICs for light weight head-stages incorporating small, efficient, and implantable light sources, light weight battery or wireless power transfer technology, and small but enhanced computational resources for long duration behavioral studies. Table III presents a summary of the state-of-the-art neural interface technology for closed loop optogenetic brain stimulation.

The light sources in some of the earliest brain stimulation experiments using optogenetics were large table top laser devices, where light is guided to the specific region of the brain using optical fibers [118], [119]. As a process of miniaturization and improving efficiency, those large light sources have been replaced by small laser diodes and μLEDs, which come in μm dimensions, thus eliminating the use of optical fibers. μLEDs have the advantages of compact size, low power consumption, and negligible heat generation and are being used in most implantable modules of the untethered optogenetic devices.

The most important attributes to be met during the design process of front end analog signal conditioning circuit and backend stimulation control circuit are PSRR, CMRR, and good SNR ratio [120]. The analog front-end circuit responsible for conditioning the sensed neural signals has also undergone a significant evolution. Some of the earlier publications we reviewed have used a commercially available digital electrophysiology interface chip offered by companies like Intan Technologies for the analog front-end stage. This interface chip is a fully integrated amplifier array with on-chip 16-bit ADC with low input noise 2.4 V_{rms}. Upper (100 Hz – 20 kHz) and lower (0.1 Hz – 500 Hz) cut off frequencies can be programmed into the chip. The package size for the chip is 8 mm × 8 mm, with the gain of up to 60 dB and supply voltage range of 3.2 - 3.6 V. Recent developments in micro/nano fabrication technology have enabled the design of neural front-end circuits of small form factor, in the configuration of head mountable or implantable chips. Some researchers have tried to further reduce the form factor of the neural interface by using CMOS technology. ASIC designs based on CMOS technology provide the advantages of increased efficiency by optimizing noise, power consumption, size, and speed of operation. Different design strategies like the use of auto-zero amplifiers to improve precision and eliminate commonly occurring offset error and flicker noise, programmable gain amplifiers and programmable bandwidth filters for sensing neural signals of particular frequencies, sigma-delta ADC for improving resolution could be observed in recent designs of the electrophysiological signal sensing circuit setup. The back-end stimulation circuit helps to generate precisely timed and high-intensity light pulses as required in optogenetic applications to activate the Channelrhodopsins and Halorhodopsins to excite and inhibit neurons, respectively. Hence, a high driving current is required, and the current should change as per the feedback

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TABLE III
EXISTING CLOSED-LOOP OPTOGENETIC BRAIN STIMULATION SYSTEMS

References	[77, 78] (2019)	[71] (2018)	[80] [76] (2020)	[72] (2018)
Portability	Tether-less	Tether-less	Tether-less	Tethered
Biomarker	LFP, AP	LFP	LFP	LFP, AP
No: of recording channels	10	16	4	32
Recording circuit	ASIC	COTS	ASIC	COTS
Gain	NG	NG	49.54 dB	NG
CMRR	NG	NG	65.88 @10 Hz	NG
Input referred noise	3.2 μV_{rms}	NG	2.08 μV_{rms}	NG
Bandwidth	0.1-7kHz	NG	0.02 – 830 Hz	NG
Power consumption	11.2 μW/channel	NG	9.8 μW/channel	NG
Algorithm	Adaptive thresholding, wavelet compression	Amplitude threshold, ON/OFF control	Amplitude threshold	Spike sorting, thresholding
Platform	ASIC + FPGA	MCU	ASIC + MCU	PC
Data transmission	Wireless	-	SPI protocol	USB interface
Light source	LED	μLED	μLED	μLED
Stimulating circuit	ASIC	COTS	ASIC	ASIC
No: of channels	4		6	12
Stimulation current (mA)			1.1	1.023
Power source	Battery	Battery	Battery	Battery
Power Consumption	56.4 μW/channel	8 mW	1.38 mW	218 μW (for the headstage)
Dimension	1.61 cm x 1.16 cm	21.2 mm x 31.8 mm x 1 mm (board)	10 mm x 10 mm (headstage) 25 mm x 22 mm (control unit)	2.16 cm x 2.38 cm x 0.35 cm
Weight (with battery)	3g	11.3g	1 g(headstage) 4.1 g (control unit)	1.9g

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signal generated by the control unit in a closed loop device. Commercially available bench top LED drivers like M470F1 by Thorlabs were used in the earlier optogenetic brain stimulation experiments. However, small (millimeter sized) LED driver chips have enabled miniaturization of the stimulation circuitry. Some publications have considered using a single application specific chip integrating the functionalities of the analog signal conditioning and stimulation circuits based on the CMOS fabrication technology and have been successful in achieving completely portable and small devices for experimentation on freely moving lab animals. However, the downside of ASIC design for the neural interface circuit is the increase in cost and complexity of the design.

ASIC with built-in digital signal processing, FPGAs, general purpose microcontrollers and external computers could be the platforms for performing the signal processing and generating

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control signals in untethered settings. Dedicated ASIC implementations can be low power provided they are implemented on a sufficiently advanced technology node. However, the downside is that they cannot be reconfigured, and hence lack flexibility which is highly desirable for the underlying platform. FPGAs, on the other hand are configurable and computationally sound and could be used to implement computationally demanding tasks such as spike sorting and control algorithms. There is always a tradeoff between computational complexity and power requirements and even with the availability of ultra-low power FPGAs, they could be more power hungry than microcontrollers. While, limited computational power of microcontrollers makes the implementation of complex spike sorting and control algorithms difficult. Wireless data communication techniques including Bluetooth technology can be used to receive user control signals from an external computer, where the complex computations are done. Including such communication protocols in the design helps in storing the incoming data to an external high capacity storage system for future offline analysis. However, such a setup has restricted range of distance of operation.

Regarding the feature extraction, classification, and control algorithms for closed loop brain stimulation, we have observed that most of the published works on miniaturized optogenetic systems have used the simple threshold detection techniques for identifying abnormalities, simple binary classification strategies for grouping the signals into either normal or abnormal category, and mainly an ON/OFF control method for controlling the stimulation light. This could be due to the fact that fast and real-time processing of brain signals puts a computational burden on resource constraint untethered system. However, for more sophisticated systems and for the realization of efficient brain-machine interfaces, research should be widened to implement classifiers based on advanced artificial neural networks and intelligent control algorithms in an untethered scenario.

Nowadays, machine learning (ML) methods have become an important tool for data analysis and control. For tasks such as anomaly detection in sensors' time-series data, most suited ML algorithms come under the category of deep neural networks (DNNs), where multiple neural layers are employed to deliver the outcome. In the case of tether-less closed loop optogenetic brain stimulation devices, there is a requirement for on-device data analysis, where the stimulation needs to be adjusted based on the analysis. Hence, deployment of the DNNs on tiny resource constraint edge devices such as FPGAs or MCUs is needed. ML algorithms on edge devices is a growing field with the availability of open source frameworks like TensorFlow Lite Micro (TFLM) [121] for running DNNs on edge devices. TFLM can be deployed on ARM Cortex–M and ESP32 MCUs that have the memory constraints. Fig. 8 presents a block diagram description of a plausible portable closed loop optogenetic brain stimulation system with a DNN classifier. The DNN classifier is trained in a TensorFlow training environment on a PC. The trained model is then converted into TFLM after optimizing for size and accuracy and deployed into an ARM Cortex-M MCU.

Also, as more medical research is concentrating on providing phase-specific stimulation to manipulate individual brain oscillation, there is a demand for developing efficiently engineered systems for meeting the challenge of rapidly analyzing the brain signals for phase estimation and generating timely stimulation on the fly.

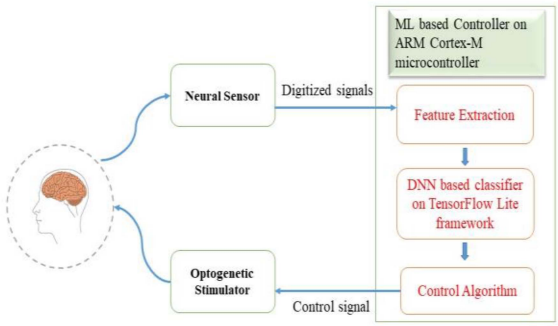


Fig. 8. Block diagram for a plausible ML-based closed loop optogenetic brain stimulation device.

Like any other tether-less device, maximizing the performance of closed-loop algorithms using the least amount of energy is the ideal requirement of untethered optogenetic brain stimulation devices. As mentioned earlier, high performance data processing demands more computational elements and memory resources. These improve the algorithm efficiency, but use more energy. Hence, there will always be a trade-off between computational power of the implanted device and its power consumption. For a tether-less device, the algorithm can be deployed on the device, or the sensed data can be sent to a computationally powerful external machine wirelessly [122]. For example, in ref. [123], wearable IoT DBS sensors are used to collect and transmit neural data to perform processing using powerful and computationally demanding heuristic tubu optimized sequence modular neural networks (HTSMNN) classification algorithms implemented in MATLAB on a PC. While passing data to the external platform helps in achieving powerful data processing, allowing the on-device processing of the data has its advantages of reducing the expenditure of the wireless transceiver.

Power management unit consists of batteries and regulators. Devices running on rechargeable batteries form the majority of existing portable closed loop optogenetic systems. Batteries account for around or even more than half of the weight of the head-stages in the optogenetic devices. Light weight batteries are to be used for reducing the weight of the device to enable ease of untethered operation without hindering the freedom of movement of the small laboratory animals. The duration over which the batteries hold charge for long term untethered operation is also an area of concern. Moreover, batteries in these implantable devices could contribute to the heat increase which could be an area of concern. With the development in wireless communication and wireless power transfer technology, controlling the stimulation through an external controller could also help in realizing untethered optogenetic simulation systems for battery-less operation. Of course, the bottleneck here is the range constraint.

Apart from these design considerations, there are other challenges to be dealt with while building a closed loop optogenetic brain stimulation system, Fig. 9 lists the issues. Possible tissue heating in in-vivo light delivery due to photothermal heating is one of the major concerns among the scientific community.

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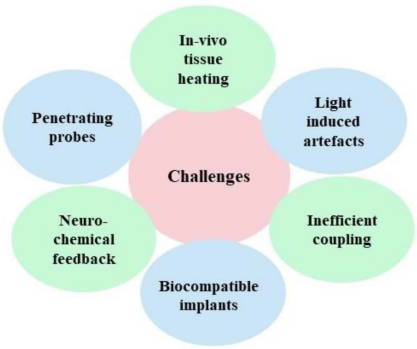


Fig. 9. Challenges to handle in the development of closed-loop optogenetic brain stimulation devices.

It is a known fact that photons in light transfer their energy to the interacting body in the form of heat. As a result of this, the temperature of the body in contact with the light increases. In the context of optogenetic brain stimulation using light, this indicates that the brain tissues getting illuminated by the light from the laser/LED optical sources can get heated up due to their exposure to light. To prevent the occurrences of adverse effects, temperature increase in the tissue surrounding the probe should be below 2 °C, as reported in some of the references [124], [125]. As temperature increase could lead to serious damage to the brain tissue, it is always good to have a temperature monitoring mechanism in place. Placing the stimulating light source external to the brain to avoid direct contact of heat source with the brain tissue can be realized in implantable devices with the help of integrated waveguides. However, it is to be considered that LEDs create light by the process called electroluminescence. Non-radiative heat is generated in the LED due to the inherent inefficiency of the electroluminescence process. This heat can be transferred to the brain tissue by the process of conduction. This heating effect becomes more distinct at higher current levels [126]. In ref. [127], McAlinden *et al.* examined the thermal properties of a μLED based probe for optogenetics and reported that the local tissue temperature rises, but to a value within the threshold limit.

Along with μLED, a laser diode can also be considered as the light source. A laser diode, even though has the advantage of coherent emission profile, poses the disadvantage of being power demanding. On the other hand, a μLED with its Lambertian emission profile causes coupling inefficiency.

Another major problem to be considered during combined optogenetic stimulation and electrophysiological recording is the light induced artefact. The close proximity of light source and recording metal electrodes on a compact scale makes the effect of light induced artefacts predominant. Such light induced artefacts distort the recordings as voltage swings, not corresponding to any electrical activity in the brain, might get recorded [128]. As a method to prevent this from happening, recently, a few closed loop optogenetic systems have made use of transparent graphene microelectrode array instead of metal electrodes to ensure that the recorded potential values are not corrupted due to effects on the electrodes by the stimulating light [69], [129], [71].

Adding to the challenges, most currently used optogenetic probes are bifurcated structures at their tip and may cause extensive tissue damage upon insertion into the brain. The minimized tissue damage may lead to obtaining neurophysiological data that have not been possible to see before. The durable tip allows many re-uses after proper cleaning and storage in between experiments. Along with this, biocompatibility of the implant material is of paramount importance for the durability of the implant. Biocompatibility is required for preventing the occurrence of chronic inflammatory events in the long term interaction with brain tissue [130]. Encapsulation technologies for implantable biomedical devices should be considered for ensuring biocompatibility and increasing safety and stability of the neural implants. Along with the conventional technologies based on ceramic and titanium packaging, other approaches like thin film coating of Al2O3, HfO2, SiO2, SiC, diamond, organic polymers of polyimide, parylene, liquid crystal polymer, silicone elastomer, SU-8, and cyclic olefin copolymer are also being studied for encapsulation of microfabricated implants [131].

It is also worth noting that all the recent reviewed works follow the concept of sensing the electrical signals from the subject to be used as the feedback information for the stimulation. However, various neurochemicals like dopamine, serotonin, histamine, taurine, etc. are proved to be closely linked with various neurological disorders and should be considered together with the electrophysiological biomarkers in realizing efficient brain stimulation systems.

X. Conclusion

The paper reviews closed loop optogenetic brain stimulation devices in detail, with an emphasis on untethered operation. The purpose of the review was to understand the evolutions in the optogenetic brain stimulation devices over the years since their first successful use. It is clear from the review that there is still ample scope for improvements in developing efficient, effective, and standalone devices. Advancements in all the three supporting aspects (genetics, optics, and electronics) can be utilized for the betterment of the closed loop optogenetic devices. From the system design point of view, integration of real-time readout of biochemical biomarkers, employing sophisticated feature extraction, classification, and control algorithm design, and systems for effectively addressing the disadvantages of temperature rise in the tissues caused by the stimulation light are understood as the possible areas of improvement. Optogenetics brain stimulation offers potential for remarkable advance in the coming decade with the help of continued extensive research in this area to improve the different facets of systems engineering by incorporating developments in micro-fabrication, sensors, computational platforms, batteries, wireless power transfer techniques, and ML.

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